



DETAILED SYSTEM ARCHITECTURE

Vendor-Agnostic DVR/NVR Forensic Analysis Platform

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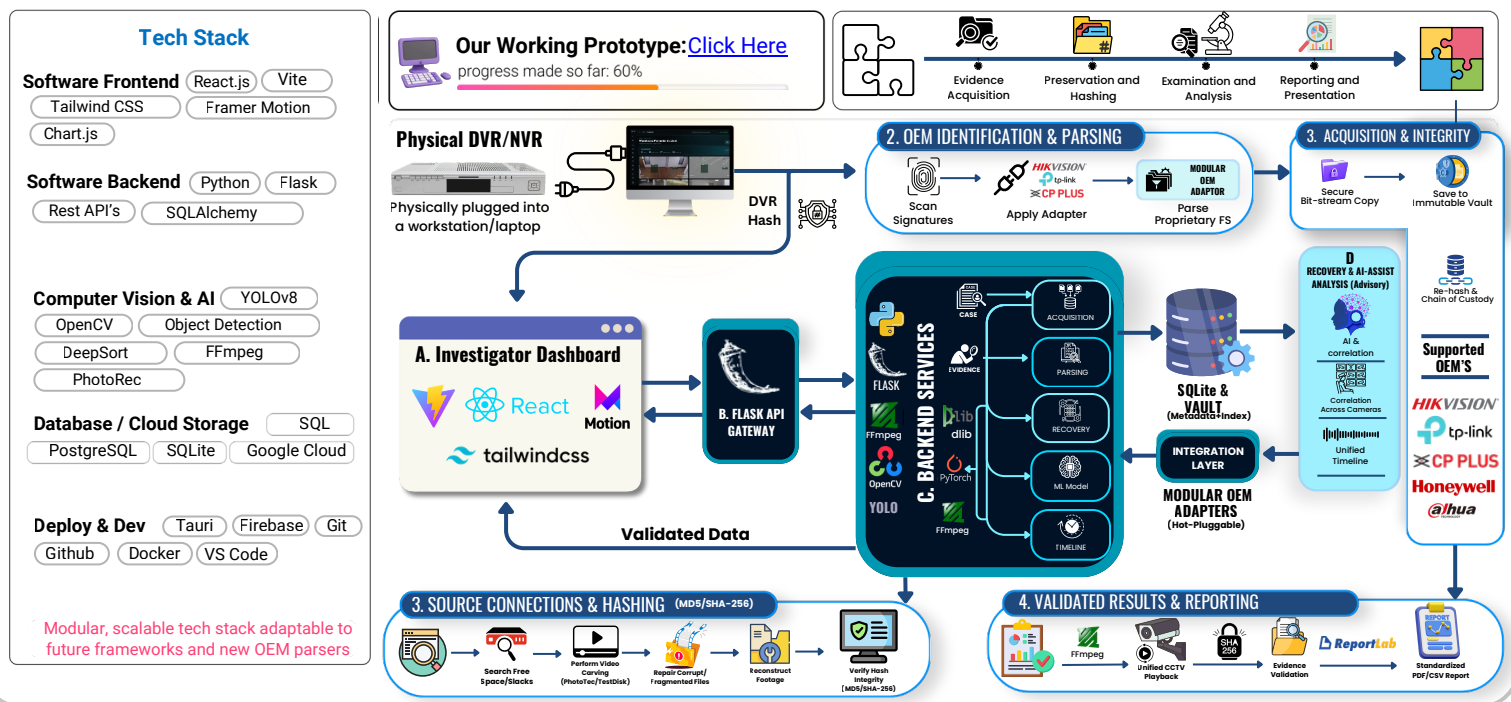


Figure 1. TRACE-X Detailed System Architecture

1. INTRODUCTION

TRACE-X follows a modular forensic workflow that connects physical DVR/NVR evidence acquisition with OEM identification, parsing, recovery, analysis, integrity verification and standardized reporting.

2. TECHNOLOGY STACK OVERVIEW

Table 1: Technologies and Tools vs Key Role in Architecture

Category	Technology and Tools	Key Role in Architecture
Frontend UI	React.js, Vite, Tailwind CSS, Framer Motion, Chart.js	Interactive Investigator Dashboard with real-time analytics and video timeline controls.
Backend	Python, Flask, REST APIs, SQLAlchemy	Authentication API Gateway microservice orchestration, secure routing, user authentication, and system logic execution.
Computer Vision & AI	YOLOv8, OpenCV, PhotoRec, FFmpeg	Object detection, continuous tracking, facial recognition, video transcoding, and automated stream repair.
Database & Storage	PostgreSQL, SQLite, Vault, Google Cloud Storage, Google Sheets	Storing structured metadata, raw evidence storage, hash registries, and system logs.
Deployment & DevOps	Git, GitHub, Firebase, Tauri, VS Code	Modular microservices deployment, version control, and cloud API hosting.

3. System Architecture and Technical Approach

TRACE-X is an end-to-end digital video forensics platform designed for NVR/DVR footage acquisition and digital forensics investigators. Unifies the entire field investigation, from initial physical media acquisition and video carving to AI-assisted analysis and alterations of free court ready reporting—into a single seamless workflow.

Stage 1: Source Connections and Hashing (Acquisition)

- Physical Interface: Software connects straight to the DVR or hard drive using a safe adapter so the original files cannot be changed or deleted.
- Integrity Baselineing: Generates a unique security code (MD5 or SHA-256) as soon as the drive is plugged in, proving the files have not been modified.

- Video Carving & Data Recovery:
 - Searches unused and leftover storage areas across the entire drive for hidden data.
 - Uses smart recovery tools (like PhotoRec and TestDisk) to automatically find lost or deleted video files.
 - Repairs corrupted or fragmented raw video file blocks to reconstruct playable footage from damaged or intentionally wiped file systems
- Verification: Re-verifies file hashes (MD5 / SHA-256) after recovery to document file state.
- Reporting Module: Generates tamper-proof, court-ready forensic reports.

Stage 2: OEM Identification and Parsing (Preservation and Hashing)

- Signature Scanning: Scans raw data streams for proprietary file system signatures and headers.
- Modular Hot-Pluggable Adapters: Utilizes plug-and-play parser modules built specifically for proprietary DVR/NVR file systems from leading OEMs (Hikvision, CP Plus, Dahua, TP-Link, Honeywell).
- Flask API Gateway Routing: Custom connector modules read the raw drive layout and route organized video stream data through our Flask server to background services.

Stage 3: Acquisition and Integrity (Examination and Analysis)

- Bit-Stream Copying: Creates a bit-level sector-by-sector clone of the original storage media to prevent working directly on original evidence.
- Immutable Vault Storage: Saves structured metadata, indices, and raw streams into a secured SQLite and Vault storage system.
- Re-hashing and Chain of Custody: Continuous automated hash re-verification maintains an immutable, court-admissible audit trail throughout analysis.

Stage 4: Validated Results and Reporting (Reporting and Presentation)

- Unified CCTV Playback: Integrates FFmpeg backend services to transcode proprietary CCTV streams into standard web formats for multi-camera video playback inside the user interface.
- Evidence Validation: Automatically validates extracted files and metadata integrity prior to report generation.
- Standardized Reporting: Generates court-admissible PDF and CSV forensic reports complete with hash verifications, timestamps, audit histories, and AI analysis findings.

4.Evidence Intelligence a AI Analytics Layer

Beyond raw file extraction, TRACE-X provides intelligent advisory analysis using computer vision models:

- Multi-Camera Timeline Normalization: Aligns multiple CCTV camera feeds across different angles and timestamps into a single, synchronized timeline.
- AI Object Detection and Tracking: Integrates YOLOv8 and OpenCV to detect persons, vehicles, and objects of interest.
- Multi-Object Tracking (DeepSORT): Tracks subjects continuously across multiple non-overlapping camera feeds.
- Facial and Feature Matching: Uses tools like dlib for targeted facial identification and ambiguity detection.

5.Key System Differentiators

- Unified Forensic Workflow: Replaces fragmented, single-purpose utilities by integrating drive carving, OEM parsing, AI analytics, and report generation into a single end-to-end interface.

- **Modular OEM Adapter Architecture:** Hot-pluggable design enables rapid integration of new proprietary OEM DVR/NVR formats without re-architecting the core backend.
- **Evidence Integrity by Design:** Enforces SHA-256 verification and automated chain-of-custody tracking across every operation to guarantee court admissibility.